

Jeffy Joseph

Union City, CA 94587

+1-(906)-275-8200 | jeffyje@gmail.com

www.linkedin.com/in/jeffjosephcfd

PROFESSIONAL EXPERIENCE

Intuitive Surgical, Mechanical Engineer

May '23 – Present

- **Robotics Hardware Expertise:** Mechanical Lead for a Class II medical device that supports advanced robotic surgery and [Da Vinci Robotic system](#), took the device from proof-of-concept through validation and into production.
- **Team Leadership:** Directed a team of **5 mechanical engineers** and partnered with ID, EE and Systems groups. Owned **70+ custom mechanical & pneumatic parts** through requirements, simulation, prototyping and validation.
- **Thermal / Vibration / Acoustic Engineering:** Architected compliant solutions that meet **IEC 60601 noise & safety targets**. Used CFD & FEA to optimise temperature, structural integrity and sound pressure levels.
- **Reliability & Compliance:** Authored the **system-level reliability plan** and designed custom fixtures to carry out essential verification and validation tests.
- **DFM/DFA & Manufacturing:** Led cost-driven DFM/DFA across injection-moulding, die-casting, sheet-metal, machining, stamping and 3-D printing, released tooling in US and O-US.
- **Cross-Functional Integration:** Co-designed PCBs, valve manifolds and pneumatics with electrical & software teams, ensuring seamless mechanical-electronic integration and regulatory readiness.

Google, Mountain View, CA

May '20 – Mar '23

Mechanical Engineer

Nov '20 – Mar '23

Program: [Nest Display](#) and [Pixel Tablet](#)

- **Mechanical and Electro-Mechanical Design:** Led mechanical design of components and subassemblies from concept through production for Google Pixel Tablet and Nest Display products.
- Designed injection molded, cast, and sheet metal parts optimized for DFM/DFA, weight reduction, and cost targets.
- Developed **thermal management solutions** balancing performance requirements, user experience, and environmental protection.
- Collaborated with EE teams to **integrate PCBAs and other electronics**, ensuring performance and thermal efficiency.
- Collaborated with the Industrial Design team to explore, prototype, and test new product concepts, as well as finalize product form and CMF.
- Developed and executed **reliability test plans** (thermal, mechanical, environmental, DoE) and performed root cause failure analysis to drive rapid design iteration.
- Worked with vendors to dial in **DFM/DFA**, quote & kick-off parts at prototype and production quantities. Owned tooling release, bring up & validation for a variety of parts & processes (Injection molding, die casting, sheet metals).
- Collaborated with JDM and CM partners to guide product concepts through rigorous validation builds (EVT, DVT, PVT) and into high-volume mass production (1M+).

Thermal Mechanical Engineer

May '20 – Nov '20

- **Loon: Thermal System Development:** Led thermal system development for stratospheric satellite electronics from concept to production, serving as both thermal and mechanical engineer.
- **CFD Analysis:** Utilized CFD analysis to architect and optimize thermal management solutions for electronic performance in extreme stratospheric conditions.
- **Waymo:** Developed early design concepts for a liquid cooling system for the third-generation long-range LIDAR, a critical component for autonomous vehicle technology. This involved close collaboration with vendors to design and develop cold plates, size fluid delivery systems, and optimize pump, radiator, and fan configurations.

Abbott Laboratories, Mechanical Engineer

Mar '19 – May '20

- **Medical Device Design:** Led the design and development of mechanical assemblies for Class II and III medical devices using SolidWorks. Responsible for driving the design of [HeartMate 3 battery charger](#) from concept to validation.
- **Thermal and Structural Analysis:** Conducted FEA and CFD analyses on biomedical devices to optimize mechanical and thermal designs.
- **Prototype Development:** Created and tested prototypes, contributing to the design of test methods for device performance evaluation.
- **Cross-Functional Teamwork:** Collaborated with product development teams to gather requirements and validate computational models.

DEKA Research and Development, Mechanical Engineer*Jan '17 – Mar '19*

- **Design and Prototyping:** Conceptualized and designed plastic and metal components using SolidWorks; evaluated prototypes with a focus on thermal performance.
- **Thermal Management Solutions:** Conducted thermal analyses and developed solutions such as heat spreader plates, heat sinks, liquid cooling, and air cooling for medical devices.
- **Process Development:** Worked with manufacturing teams to establish assembly processes and instructions, ensuring thermal considerations were integrated.

TECHNICAL SKILLS**Mechanical Design Software:** NX, SolidWorks, 3DCS, Pro-E**Mechanical Engineering:** GD&T, DFM/DFA, Thermal Management, Structural Analysis, Prototyping**Thermal and CFD Analysis:** ANSYS Fluent, ICEPAK, FloTHERM, OpenFOAM, COMSOL Multiphysics, Paraview**Programming and Tools:** Python, MATLAB, C++

EDUCATION**Master of Science in Mechanical Engineering***Aug '15 – Apr '17*

- Michigan Technological University, Houghton

*GPA: 3.85/4.0***Bachelor of Technology in Mechanical Engineering***Aug '10 – May '14*

- Mahatma Gandhi University

GPA: 7.47/10.0